

NT-C-032 | Cyber Security | Course Knowledge Profile

Identity, purpose, prerequisites, scope and boundaries

Category	Cybersecurity
Target learners	Learners targeting entry-level security operations and support roles.
Prerequisites	Networking, operating-system and Linux fundamentals are recommended.
Approved duration	16-20 weeks / 180-240 guided hours.
Final outcome	Identify, monitor and respond to common security risks using authorised labs and documented controls.
Assessment profile	Infrastructure/Security

Approved tools / platforms

Authorised cyber lab, Windows/Linux, network tools, vulnerability scanner, log/SIEM platform and documentation tools.

Knowledge boundaries

Include	Quality rule	Exclude / escalate
The eight approved modules, four sections, named tools, projects and mapped entry-level roles.	Require least privilege, authorised labs, configuration evidence, logs, monitoring, rollback and documented troubleshooting.	Never recommend unsafe production testing or treat a scanner output as a verified finding without validation.
Current product/platform procedures only when version and date are known.	Use primary documentation and record the source version.	Do not invent current commands, prices, tax rates, policies or certification status.
Evidence-based career guidance.	Cite tests, projects, capstone, viva and verified resume evidence.	No guaranteed job, placement, salary, interview or employer claim.

LLM answer pattern: identify the course and version, answer from approved records, cite record IDs, distinguish verified facts from guidance, and state any missing evidence.

NT-C-032 | Cyber Security | NT-C-032-S01 Deep Knowledge

Modules 1-2 | objectives, concepts, evidence and remediation

Section competency	CIA, threats, controls, risk and ethics; Protocols, segmentation, firewalls and secure access
Completion evidence	Risk worksheet; Network-control lab
Section weight	20% of the weighted mock-test average
Assessment	10 MCQ (30 marks) + 4 troubleshooting scenarios (30) + 1 authorised practical lab (40)

Module 1 - Security foundation

Objective: CIA, threats, controls, risk and ethics

Observable evidence: Risk worksheet

Concept	Approved knowledge statement	Application behaviour	Evidence
CIA	A defined competency within Cyber Security. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when cia is appropriate; apply it in a Cyber Security task; verify the result.	Risk worksheet
threats	A defined competency within Cyber Security. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when threats is appropriate; apply it in a Cyber Security task; verify the result.	Risk worksheet
controls	A defined competency within Cyber Security. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when controls is appropriate; apply it in a Cyber Security task; verify the result.	Risk worksheet
risk	A defined competency within Cyber Security. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when risk is appropriate; apply it in a Cyber Security task; verify the result.	Risk worksheet
ethics	A defined competency within Cyber Security. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when ethics is appropriate; apply it in a Cyber Security task; verify the result.	Risk worksheet

Module 2 - Network security

Objective: Protocols, segmentation, firewalls and secure access

Observable evidence: Network-control lab

Concept	Approved knowledge statement	Application behaviour	Evidence
Protocols	A defined competency within Cyber Security. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when protocols is appropriate; apply it in a Cyber Security task; verify the result.	Network-control lab
segmentation	A defined competency within Cyber Security. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when segmentation is appropriate; apply it in a Cyber Security task; verify the result.	Network-control lab
firewalls	A control that allows or blocks traffic according to defined rules. In Cyber Security, it must be demonstrated through an observable task, not only recalled as a definition.	Explain when firewalls is appropriate; apply it in a Cyber Security task; verify the result.	Network-control lab
secure access	A defined competency within Cyber Security. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when secure access is appropriate; apply it in a Cyber Security task; verify the result.	Network-control lab

Remediation: If the learner cannot explain the purpose, complete the stated task or provide Risk worksheet; Network-control lab, return to Modules 1-2, complete one guided practice and then use a new equivalent test form.

NT-C-032 | Cyber Security | NT-C-032-S02 Deep Knowledge

Modules 3-4 | objectives, concepts, evidence and remediation

Section competency	Windows/Linux hardening, patching and baselines; Authentication, authorization, MFA and privilege
Completion evidence	Hardened host; IAM review
Section weight	25% of the weighted mock-test average
Assessment	10 MCQ (30 marks) + 4 troubleshooting scenarios (30) + 1 authorised practical lab (40)

Module 3 - System security

Objective: Windows/Linux hardening, patching and baselines

Observable evidence: Hardened host

Concept	Approved knowledge statement	Application behaviour	Evidence
Windows/Linux hardening	A defined competency within Cyber Security. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when windows/linux hardening is appropriate; apply it in a Cyber Security task; verify the result.	Hardened host
patching	A defined competency within Cyber Security. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when patching is appropriate; apply it in a Cyber Security task; verify the result.	Hardened host
baselines	A defined competency within Cyber Security. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when baselines is appropriate; apply it in a Cyber Security task; verify the result.	Hardened host

Module 4 - Identity

Objective: Authentication, authorization, MFA and privilege

Observable evidence: IAM review

Concept	Approved knowledge statement	Application behaviour	Evidence
Authentication	Verification of identity. In Cyber Security, it must be demonstrated through an observable task, not only recalled as a definition.	Explain when authentication is appropriate; apply it in a Cyber Security task; verify the result.	IAM review
authorization	Control of what an authenticated identity is allowed to do. In Cyber Security, it must be demonstrated through an observable task, not only recalled as a definition.	Explain when authorization is appropriate; apply it in a Cyber Security task; verify the result.	IAM review
MFA	A defined competency within Cyber Security. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when mfa is appropriate; apply it in a Cyber Security task; verify the result.	IAM review
privilege	A defined competency within Cyber Security. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when privilege is appropriate; apply it in a Cyber Security task; verify the result.	IAM review

Remediation: If the learner cannot explain the purpose, complete the stated task or provide Hardened host; IAM review, return to Modules 3-4, complete one guided practice and then use a new equivalent test form.

NT-C-032 | Cyber Security | NT-C-032-S03 Deep Knowledge

Modules 5-6 | objectives, concepts, evidence and remediation

Section competency	Asset inventory, scanning, prioritisation and remediation; Logs, SIEM concepts, alerts and triage
Completion evidence	Vulnerability report; Detection lab
Section weight	25% of the weighted mock-test average
Assessment	10 MCQ (30 marks) + 4 troubleshooting scenarios (30) + 1 authorised practical lab (40)

Module 5 - Vulnerability management

Objective: Asset inventory, scanning, prioritisation and remediation

Observable evidence: Vulnerability report

Concept	Approved knowledge statement	Application behaviour	Evidence
Asset inventory	A defined competency within Cyber Security. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when asset inventory is appropriate; apply it in a Cyber Security task; verify the result.	Vulnerability report
scanning	A defined competency within Cyber Security. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when scanning is appropriate; apply it in a Cyber Security task; verify the result.	Vulnerability report
prioritisation	A defined competency within Cyber Security. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when prioritisation is appropriate; apply it in a Cyber Security task; verify the result.	Vulnerability report
remediation	A defined competency within Cyber Security. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when remediation is appropriate; apply it in a Cyber Security task; verify the result.	Vulnerability report

Module 6 - Security monitoring

Objective: Logs, SIEM concepts, alerts and triage

Observable evidence: Detection lab

Concept	Approved knowledge statement	Application behaviour	Evidence
Logs	A defined competency within Cyber Security. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when logs is appropriate; apply it in a Cyber Security task; verify the result.	Detection lab
SIEM concepts	A defined competency within Cyber Security. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when siem concepts is appropriate; apply it in a Cyber Security task; verify the result.	Detection lab
alerts	A defined competency within Cyber Security. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when alerts is appropriate; apply it in a Cyber Security task; verify the result.	Detection lab
triage	A defined competency within Cyber Security. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when triage is appropriate; apply it in a Cyber Security task; verify the result.	Detection lab

Remediation: If the learner cannot explain the purpose, complete the stated task or provide Vulnerability report; Detection lab, return to Modules 5-6, complete one guided practice and then use a new equivalent test form.

NT-C-032 | Cyber Security | NT-C-032-S04 Deep Knowledge

Modules 7-8 | objectives, concepts, evidence and remediation

Section competency	Preparation, containment, evidence and recovery; Policies, cloud basics, awareness and reporting
Completion evidence	Incident playbook; Security operations case
Section weight	30% of the weighted mock-test average
Assessment	10 MCQ (30 marks) + 4 troubleshooting scenarios (30) + 1 authorised practical lab (40)

Module 7 - Incident response

Objective: Preparation, containment, evidence and recovery

Observable evidence: Incident playbook

Concept	Approved knowledge statement	Application behaviour	Evidence
Preparation	A defined competency within Cyber Security. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when preparation is appropriate; apply it in a Cyber Security task; verify the result.	Incident playbook
containment	A defined competency within Cyber Security. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when containment is appropriate; apply it in a Cyber Security task; verify the result.	Incident playbook
evidence	A defined competency within Cyber Security. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when evidence is appropriate; apply it in a Cyber Security task; verify the result.	Incident playbook
recovery	A defined competency within Cyber Security. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when recovery is appropriate; apply it in a Cyber Security task; verify the result.	Incident playbook

Module 8 - Governance capstone

Objective: Policies, cloud basics, awareness and reporting

Observable evidence: Security operations case

Concept	Approved knowledge statement	Application behaviour	Evidence
Policies	A defined competency within Cyber Security. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when policies is appropriate; apply it in a Cyber Security task; verify the result.	Security operations case
cloud basics	A defined competency within Cyber Security. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when cloud basics is appropriate; apply it in a Cyber Security task; verify the result.	Security operations case
awareness	A defined competency within Cyber Security. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when awareness is appropriate; apply it in a Cyber Security task; verify the result.	Security operations case
reporting	A defined competency within Cyber Security. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when reporting is appropriate; apply it in a Cyber Security task; verify the result.	Security operations case

Remediation: If the learner cannot explain the purpose, complete the stated task or provide Incident playbook; Security operations case, return to Modules 7-8, complete one guided practice and then use a new equivalent test form.

NT-C-032 | Cyber Security | Skill Ontology & Dependency Map

Atomic skills, learning order and evidence strength

Skill ID	Competency	Depends on	Evidence	Type
NT-C-032-SK01	Security foundation	Prerequisite foundation	Risk worksheet	Critical
NT-C-032-SK02	Network security	M01 plus earlier foundations	Network-control lab	Critical
NT-C-032-SK03	System security	M02 plus earlier foundations	Hardened host	Supporting
NT-C-032-SK04	Identity	M03 plus earlier foundations	IAM review	Supporting
NT-C-032-SK05	Vulnerability management	M04 plus earlier foundations	Vulnerability report	Supporting
NT-C-032-SK06	Security monitoring	M05 plus earlier foundations	Detection lab	Critical
NT-C-032-SK07	Incident response	M06 plus earlier foundations	Incident playbook	Supporting
NT-C-032-SK08	Governance capstone	M07 plus earlier foundations	Security operations case	Critical

Evidence-strength hierarchy

Strength	Evidence source	Use
High	Trainer-observed practical, viva, working source/project files and reproducible output.	May support verified mastery when rubric threshold is met.
Medium	Scored mock tests, reviewed assignments and documented portfolio artifacts.	Supports mastery with corroboration.
Low	Resume claim, self-report, attendance or video completion without demonstration.	May trigger verification; cannot alone support Apply Now.

CareerTours must not treat attendance, time spent or content opened as skill mastery.

NT-C-032 | Cyber Security | Assessment Knowledge & Sample Items

Non-live examples for LLM training and evaluation

Sample ID	Type	Question	Approved answer / rubric anchor
NT-C-032-S01-EX01	Evidence/application	What evidence best demonstrates the combined competency of Security foundation and Network security?	The learner should provide Risk worksheet; Network-control lab and explain how the work applies CIA, threats, controls, risk and ethics; Protocols, segmentation, firewalls and secure access. Recall alone is insufficient.
NT-C-032-S01-EX02	Scenario/remediation	A learner reports completing Network security but cannot produce Network-control lab. What should CareerTours do?	Mark the evidence Not Yet Verified, assign practice from NT-C-032-S01, request Network-control lab, and allow a new equivalent test only after remediation.
NT-C-032-S02-EX01	Evidence/application	What evidence best demonstrates the combined competency of System security and Identity?	The learner should provide Hardened host; IAM review and explain how the work applies Windows/Linux hardening, patching and baselines; Authentication, authorization, MFA and privilege. Recall alone is insufficient.
NT-C-032-S02-EX02	Scenario/remediation	A learner reports completing Identity but cannot produce IAM review. What should CareerTours do?	Mark the evidence Not Yet Verified, assign practice from NT-C-032-S02, request IAM review, and allow a new equivalent test only after remediation.
NT-C-032-S03-EX01	Evidence/application	What evidence best demonstrates the combined competency of Vulnerability management and Security monitoring?	The learner should provide Vulnerability report; Detection lab and explain how the work applies Asset inventory, scanning, prioritisation and remediation; Logs, SIEM concepts, alerts and triage. Recall alone is insufficient.
NT-C-032-S03-EX02	Scenario/remediation	A learner reports completing Security monitoring but cannot produce Detection lab. What should CareerTours do?	Mark the evidence Not Yet Verified, assign practice from NT-C-032-S03, request Detection lab, and allow a new equivalent test only after remediation.
NT-C-032-S04-EX01	Evidence/application	What evidence best demonstrates the combined competency of Incident response and Governance capstone?	The learner should provide Incident playbook; Security operations case and explain how the work applies Preparation, containment, evidence and recovery; Policies, cloud basics, awareness and reporting. Recall alone is insufficient.
NT-C-032-S04-EX02	Scenario/remediation	A learner reports completing Governance capstone but cannot produce Security operations case. What should CareerTours do?	Mark the evidence Not Yet Verified, assign practice from NT-C-032-S04, request Security operations case, and allow a new equivalent test only after remediation.

Live bank rule: Generate at least three times the live quantity for each section using the approved Infrastructure/Security blueprint. Human-review every scored item before initial production release.

NT-C-032 | Cyber Security | Labs, Projects & Scoring Rubrics

Practical proof of learning and ownership

Level	Approved project	Skills evidenced	Required deliverables
Mini 1	Hardening baseline	Host and identity controls	Brief; working/source file or reproducible environment; output; validation notes; reflection.
Mini 2	Alert triage case	Logs, detection and response	Brief; working/source file or reproducible environment; output; validation notes; reflection.
Capstone	Security operations assessment	Risk, remediation and incident plan	Brief; working/source file or reproducible environment; output; validation notes; reflection.

Standard project rubric

Dimension	Weight	What earns credit
Problem framing	15%	Clear objective, user/business need, assumptions and success criteria.
Technical/design accuracy	25%	Correct methods, appropriate tools and sound decisions.
Completeness	15%	Required features, assets, data or workflow are present.
Testing and validation	15%	Important normal, edge, quality or failure cases are checked.
Evidence and reproducibility	15%	Source/working files, setup, inputs, outputs and version information are available.
Documentation	5%	Concise structure, instructions and limitations.
Viva / ownership	10%	Learner can explain decisions, demonstrate work and respond to a change request.

Require least privilege, authorised labs, configuration evidence, logs, monitoring, rollback and documented troubleshooting.

NT-C-032 | Cyber Security | Career Roles, Evidence & Gap Actions

Explainable top-three occupation matching

Rank	Occupation	Default weighted skills	Minimum evidence	Gap action
1	SOC Analyst Trainee	Logs 34% [critical]; alerts 33% [critical]; triage 33%	Detection lab	If a critical skill is below 60 or detection lab is missing, return Improve First or Needs Evidence with the linked module/project.
2	Cybersecurity Support Associate	Hardening 50% [critical]; vulnerability management 50% [critical]	Remediation report	If a critical skill is below 60 or remediation report is missing, return Improve First or Needs Evidence with the linked module/project.
3	Junior Security Operations Analyst	Monitoring 50% [critical]; incident process 50% [critical]	Operations capstone	If a critical skill is below 60 or operations capstone is missing, return Improve First or Needs Evidence with the linked module/project.

Course readiness	45% section mocks + 20% mini projects + 25% capstone + 10% viva.
Role readiness	45% role-relevant mastery + 25% projects/capstone + 20% aligned mocks + 10% verified resume evidence.
Evidence confidence	Show separately; Apply Now requires at least 70.
Resume extraction	Use only relevant skills, projects, education, experience and portfolio evidence; minimise personal data before external model processing.

A student may be Apply Now for one mapped occupation and Improve First for another. Return independent evidence and gaps for each role.

NT-C-032 | Cyber Security | FAQs, Misconceptions & LLM Response Pack

Course-specific retrieval and response patterns

Approved FAQ	Answer
Who is Cyber Security for?	Learners targeting entry-level security operations and support roles.
What should I know before starting?	Networking, operating-system and Linux fundamentals are recommended.
How long is the approved learning journey?	16-20 weeks / 180-240 guided hours.
What will I be able to demonstrate?	Identify, monitor and respond to common security risks using authorised labs and documented controls.
Does completion guarantee a job?	No. Completion and role readiness are evidence-based guidance; employment, placement, salary and employer selection are not guaranteed.
What if I fail a section?	CareerTours identifies the failed skill tags and module, assigns targeted practice, and permits an equivalent retest after remediation.

Misconception	Approved correction
Attendance proves mastery	False - require tests and observable practical/project evidence.
One total score proves every role fit	False - calculate each occupation using its weighted skills and evidence.
The LLM can add missing details	False - retrieve approved records or state that evidence is insufficient.
Course completion guarantees employment	False - never claim guaranteed job, placement, salary or interview.
Cyber Security knowledge never changes	False - version-sensitive tools, laws, taxes, security and platform procedures require review.

Retrieval metadata

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course_code=NT-C-032; corpus_version=3.0;
record_types=course_profile,module,section,skill,assessment_sample,project,occupation_mapping,faq,misconception;
approval_status=approved; effective_date=2026-08-12
```